Graphs covered by vertex and edge identity Seidel switches

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Abstract

We study finite simple graphs in which every vertex belongs to an identity Seidel switch consisting of a singleton or the endpoints of an edge. We prove that every such graph of order $n \geq 3$ is connected and has minimum and maximum degrees satisfying $\delta + \Delta = n$. Consequently the property is incompatible with complementation and with self-complementarity. We characterize the bipartite members as the complete bipartite graphs and give an exact local characterization of the regular members, together with infinite constructions. An exhaustive enumeration through order eight gives respectively $1, 3, 3, 11, 10, 35$ isomorphism classes, with explicit certificates for every successful singleton and edge switch. A three-vertex example shows that identity switches need not be closed under composition.

1 Switches and the cut identity

Let $G = (V, E)$ be a finite simple graph of order n. For $S \subseteq V$, the Seidel switch G^S complements precisely the adjacencies between S and $V \setminus S$. Following Gervacio [1], S is an *identity Seidel switch* (ISS) if $G^S \cong G$. A singleton ISS is a vertex-ISS; an edge-ISS is an ISS $\{x, y\}$ with $xy \in E$. We call G *covered* if every vertex belongs to a vertex-ISS or an edge-ISS. All isomorphisms below may permute the entire vertex set.

Write $e_G(S, V \setminus S)$ for the number of edges crossing the cut.

Proposition 1. *For every $S \subseteq V$,*

$$|E(G^S)| - |E(G)| = |S|(n - |S|) - 2e_G(S, V \setminus S).$$

In particular, a vertex-ISS v satisfies $2\deg_G(v) = n-1$, and an edge-ISS xy satisfies $\deg_G(x)+\deg_G(y) = n$.

Proof. There are $|S|(n-|S|)$ potential cut edges. Switching removes the $e_G(S, V \setminus S)$ present ones and adds the remaining ones; all other edges are unchanged. Isomorphism implies equality of edge counts. For $S = \{v\}$ the cut has size $\deg_G(v)$. For an edge xy and $S = \{x, y\}$ its size is $\deg_G(x) + \deg_G(y) - 2$, so the identity gives $2(n-2) = 2(\deg_G(x) + \deg_G(y) - 2)$. □

For a nonadjacent pair the corresponding necessary degree sum is $n - 2$. The adjacency requirement in the definition of edge-ISS is therefore substantive.

Proposition 2. *A nonempty graph has every vertex a vertex-ISS if and only if $n = 1$.*

Proof. The singleton case is immediate. Otherwise Proposition 1 makes G regular of degree $d = (n-1)/2 > 0$. Choose an edge vw. Switching at v lowers the degree of w to $d - 1$, whereas every graph isomorphic to G is d-regular. This is a contradiction. □

This answers Problem 6.1 of [1] and explains the need to allow edge-ISS in the covering question.

2 Restrictions and two characterizations

Theorem 3. *If G is covered and $n \geq 3$, then*

$$\delta(G) + \Delta(G) = n.$$

Moreover, G is connected, its complement is not covered, and G is not self-complementary.

Proof. Put $h = (n-1)/2$, $\delta = \delta(G)$ and $\Delta = \Delta(G)$. Any vertex whose degree differs from h must belong to an edge-ISS, and thus has a neighbor of degree $n - \deg_G(v)$. If $\Delta \leq h$, a vertex of degree less than h would require a neighbor of degree greater than h. Thus all vertices would have degree h. No edge could then be an edge-ISS, since its degree sum would be $n-1$. Coverage would make all vertices vertex-ISS, contradicting Proposition 2. Therefore $\Delta > h$.

A maximum-degree vertex belongs to an edge-ISS, giving $\delta \leq n - \Delta$. If $\delta \neq h$, a minimum-degree vertex also belongs to an edge-ISS, giving $n - \delta \leq \Delta$. If $\delta = h$, then n is odd and the integer $\Delta > h$ satisfies $\Delta \geq h + 1 = n - \delta$. In both cases $\delta + \Delta = n$.

If G were disconnected, the component containing a maximum-degree vertex would have at least $\Delta + 1$ vertices and any other component at least $\delta + 1$. Their total would exceed n. Finally,

$$\delta(\overline{G}) + \Delta(\overline{G}) = 2(n-1) - (\delta + \Delta) = n - 2,$$

so $\overline{G}$ cannot be covered. Since coverage is invariant under isomorphism, G cannot be self-complementary. □

Call adjacent vertices x, y *complementary outside the pair* if every other vertex is adjacent to exactly one of them.

Theorem 4. *A regular graph G of order $n \geq 3$ is covered if and only if n is even, its degree is $n/2$, and every vertex has an adjacent vertex complementary outside the pair. In a regular covered graph these pairs are exactly the edge-ISS.*

Proof. For a regular covered graph Theorem 3 gives $2d = n$. There are no vertex-ISS, by Proposition 1. Suppose xy is an edge-ISS. For $w \notin \{x, y\}$, let c_w be the number of its neighbors in $\{x, y\}$. After switching, its degree is $d + 2 - 2c_w$. The switched graph must still be d-regular, so $c_w = 1$. Conversely, whenever x, y are complementary outside the pair, the transposition $(x\ y)$, fixing every other vertex, is an isomorphism $G^{\{x,y\}} \to G$. Such pairs covering all vertices therefore imply coverage. This sufficiency holds without regularity. □

Theorem 5. *The bipartite covered graphs of order $n \geq 3$ are exactly $K_{a,b}$ with $a, b \geq 1$ and $a + b = n$.*

Proof. By Theorem 3, a bipartite covered graph has a bipartition $A \cup B$ with $1 \leq a = |A| \leq b = |B|$. An edge-ISS xy with $x \in A$, $y \in B$ has degree sum $a + b$, so the bounds $\deg(x) \leq b$ and $\deg(y) \leq a$ are both attained. Take any $v \in B$. If it belongs to an edge-ISS, it has degree a. Otherwise it is a vertex-ISS, of degree $(n-1)/2 \leq a \leq \lfloor n/2 \rfloor$; this is possible only for odd n and $a = (n-1)/2$, again giving degree a. Every vertex in B is therefore adjacent to all of A. Conversely, every cross-part pair in $K_{a,b}$ is complementary outside the pair, so every edge is an edge-ISS. □

3 Constructions and the irregular case

For each $m \geq 2$, partition $2m$ vertices into adjacent pairs $\{a_i, b_i\}$, $1 \leq i \leq m$. Between every two pairs choose either of the two perfect matchings, independently. Each vertex has degree

m, and a_i, b_i are complementary outside their pair. This constructs regular covered graphs of every even order at least four. Choices include $K_{m,m}$ and the graph consisting of two copies of K_m joined by a perfect matching. The latter gives the triangular prism when $m = 3$.

More generally, suppose every vertex of a graph has a neighbor complementary outside the pair. Replace each vertex u by a nonempty independent set I_u, with all edges between I_u and I_v precisely when uv was an edge. For complementary adjacent u, v, every pair $x \in I_u$, $y \in I_v$ is again complementary outside the pair: other vertices in these two sets see exactly one endpoint, and all remaining sets inherit the original property. Thus arbitrary positive independent-set blow-ups preserve this sufficient condition and give irregular covered graphs as well.

The sufficient condition is not necessary for irregular graphs. The paw, with edges $03, 12, 13, 23$, is covered by $\{0, 3\}$ and $\{1, 2\}$; the transposition (0 3) certifies both switches. The pair $1, 2$ has both a common neighbor and a common nonneighbor, and neither of these vertices has a complementary neighbor. The house, with edges $01, 03, 04, 12, 23, 34$, needs a singleton in its cover: its successful switches are exactly $\{4\}$, $\{0, 1\}$ and $\{2, 3\}$.

Theorems 4 and 5 characterize two subclasses for all orders. A characterization of the remaining irregular, nonbipartite graphs remains open. The extremal degree identity alone does not characterize coverage. For example, take $K_{1,4}$ with center 4 and add the edge 01. Its degree sequence is $1, 1, 2, 2, 4$, so $\delta + \Delta = 5$. The vertex 0 has no neighbor of the required degree 3 for an edge-ISS. Switching at 0 changes the degree sequence to $1, 2, 2, 2, 3$, so 0 is not a vertex-ISS either.

4 Certified enumeration

We tested every singleton and every adjacent pair in every simple graph up to isomorphism for $3 \leq n \leq 8$. The input consists of the Read–Wilson graph atlas [2], as distributed by NetworkX for orders at most seven, and McKay's catalogue [3] at order eight. The results are:

n	All graphs	Switches tested	Covered	Regular	Bipartite
3	4	18	1	0	1
4	11	77	3	1	2
5	34	340	3	0	2
6	156	2106	11	2	3
7	1044	18270	10	0	3
8	12346	271612	35	3	4

At orders three through five the members are $K_{1,2}$; $K_{1,3}$, the paw and C_4; and $K_{1,4}$, the house and $K_{2,3}$, respectively. The regular members of order six are $K_{3,3}$ and the triangular prism. At orders five and seven, respectively one and six covered graphs have a vertex belonging to no edge-ISS, so the singleton option enlarges the class.

For each input graph the computation records a decision for every tested switch. The cut identity and degree multiset give exact rejection tests; remaining cases use exhaustive graph isomorphism. Every successful switch has a stored permutation p, with the convention

$$uv \in E(G^S) \quad \Longleftrightarrow \quad p(u)p(v) \in E(G).$$

The accompanying data include graph6 representatives, edge lists and all such certificates for the covered graphs. A separate standard-library checker decodes the graphs, toggles the cut and independently searches all compatible bijections for every tested switch. It also checks every certificate and verifies pairwise nonisomorphism of the input graphs. Catalogue

completeness through order five is additionally checked by matching every labelled graph to a representative. For orders six through eight, completeness uses the cited catalogues; the distinct representative counts agree with the established counts.

5 Composition and source distinctions

Let G have vertices $0, 1, 2$ and the sole edge 12. Switching at 1 gives the sole edge 01, with certificate (0 2); switching at 2 gives the sole edge 02, with certificate (0 1). Both singletons are ISS. Their composition is the switch by $\{1, 2\}$, which gives K_3 and hence is not an ISS. Thus the ISS family need not be closed under symmetric difference, even modulo complementary switch sets. No smaller graph has this failure: for order one all switches are trivial, and for order two a singleton changes the edge count.

The switch operators themselves do satisfy $(G^S)^T = G^{S \triangle T}$. The failure arises because an isomorphism witnessing $G^S \cong G$ may move the labels of T. The first version of [1] asserts that all ISS form a group; that assertion is false, and no such assertion is used here.

There is a further distinction relevant to structural characterization. The paw above contradicts the neighborhood-partition assertion of Lemma 5.2 in that version. Nor can an isomorphism always be chosen to exchange the switched endpoints and preserve their complement, as used in Theorem 5.3 there. For example, take the graph with edges

$$01, 03, 05, 12, 23, 24, 25, 34, 45.$$

The switch by $\{3, 4\}$ has certificate $p = (3, 2, 1, 5, 4, 0)$, where the tuple lists the images of $0, \ldots, 5$. On the remaining vertices the induced graph is the cycle $0, 1, 2, 5, 0$. The neighborhood of 3 there is $\{0, 2\}$, a nonedge, whereas the complement of the neighborhood of 4 there is $\{0, 1\}$, an edge. No automorphism of this cycle maps the first set to the second. The cut proof and the unrestricted isomorphism searches above avoid these assumptions.

References

[1] S. V. Gervacio, *On identity Seidel switches*, arXiv:2601.04530v1, 2026. `https://arxiv.org/abs/2601.04530v1`.

[2] R. C. Read and R. J. Wilson, *An Atlas of Graphs*, Oxford University Press, 1998.

[3] B. D. McKay, *Graphs*, small graph data catalogue, `https://users.cecs.anu.edu.au/~bdm/data/graphs.html`.